Assignment 12

RESTAPI in Java
Question:

Problem Statement:

For this assignment, instead of using Postman, call the servlet using Java, keeping the following things in mind:

- You must use Apache HttpClient
- You must use a form of REST Authentication to authenticate the caller/client (modify your servlet accordingly)
- The file must be uploaded using multipart
- All file types must be saved without corruption
- Upon encountering a name conflict, you must save the new file with a timestamp in the name

Repo link : Pull requests — RESTAssignment — Bitbucket

Client-side file upload

```
package org.example;
import org.apache.http.client.methods.CloseableHttpResponse;
import org.apache.http.client.methods.HttpPost;
import org.apache.http.entity.mime.MultipartEntityBuilder;
import org.apache.http.entity.ContentType;
import org.apache.http.impl.client.CloseableHttpClient;
import org.apache.http.impl.client.HttpClients;
import org.apache.http.util.EntityUtils;
import java.io.File;
import java.nio.charset.StandardCharsets;
import java.util.Base64;
public class ClientUploadsFile {
    public static void main(String[] args) {
        String servletUrl = "http://localhost:8080/myServletAssignment/hello";
        File fileToUpload = new File("C:/Users/Axeno/Downloads/sample (1).json"); //
change to your file
        try (CloseableHttpClient client = HttpClients.createDefault()) {
            HttpPost post = new HttpPost(servletUrl);
            String username = "admin";
            String password = "secret";
            String auth = username + ":" + password;
            String encodedAuth =
Base64.getEncoder().encodeToString(auth.getBytes(StandardCharsets.UTF_8));
            post.setHeader("Authorization", "Basic " + encodedAuth);
```

```
            MultipartEntityBuilder builder = MultipartEntityBuilder.create();
            builder.addBinaryBody("file", fileToUpload, ContentType.DEFAULT_BINARY,
fileToUpload.getName());
            post.setEntity(builder.build());
            try (CloseableHttpResponse response = client.execute(post)) {
                int status = response.getStatusLine().getStatusCode();
                String responseBody = EntityUtils.toString(response.getEntity());
                System.out.println("HTTP Status: " + status);
                System.out.println("Response Body:\n" + responseBody);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Servlet

```
package org.example.myservletassignment;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.MultipartConfig;
import jakarta.servlet.http.*;
import java.io.*;
import java.nio.file.Files;
import java.time.LocalDateTime;
import java.time.format.DateTimeFormatter;
import java.util.Base64;
@MultipartConfig
public class HelloServlet extends HttpServlet {
    private static final String USERNAME = "admin";
    private static final String PASSWORD = "secret";
    @Override
    public void init() {
        System.out.println("Servlet is initialized");
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
IOException {
        response.setContentType("text/plain");
        response.getWriter().println("Welcome to my assignment servlet !");
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws IOException {
        response.setContentType("text/plain");
        response.setCharacterEncoding("UTF-8");
        if (!isAuthenticated(request)) {
            response.setStatus(HttpServletResponse.SC_UNAUTHORIZED);
            response.getWriter().println("Unauthorized");
            return;
```

```
        }
        String uploadDir =
"C:/Users/Axeno/IdeaProjects/myServletAssignment/src/main/resources/uploads";
        File dir = new File(uploadDir);
        if (!dir.exists()) dir.mkdirs();
        try {
            Part filePart = request.getPart("file");
            String submittedFileName = filePart.getSubmittedFileName();
            File targetFile = new File(uploadDir, submittedFileName);
            if (targetFile.exists()) {
                String timestamp = LocalDateTime.now()
                        .format(DateTimeFormatter.ofPattern("yyyyMMddHHmmss"));
                targetFile = new File(uploadDir, timestamp + "_" + submittedFileName);
            }
            try (InputStream input = filePart.getInputStream()) {
                Files.copy(input, targetFile.toPath());
            }
            PrintWriter out = response.getWriter();
            out.println("File uploaded successfully!");
            out.println("Saved as: " + targetFile.getAbsolutePath());
        } catch (Exception e) {
            response.getWriter().println("Upload failed: " + e.getMessage());
        }
    }
    private boolean isAuthenticated(HttpServletRequest request) {
        String authHeader = request.getHeader("Authorization");
        if (authHeader == null || !authHeader.startsWith("Basic ")) {
            return false;
        }
        String base64Credentials = authHeader.substring("Basic ".length());
        String credentials = new String(Base64.getDecoder().decode(base64Credentials));
        String[] values = credentials.split(":", 2);
        return values.length == 2 &&
                USERNAME.equals(values[0]) &&
                PASSWORD.equals(values[1]);
    }
    @Override
    public void destroy() {
        System.out.println("Servlet Destroyed");
    }
}
```

Code snippet

```java
package org.example.myservletassignment;
import jakarta.servlet.http.*;
import java.io.IOException;
import java.io.PrintWriter;
import java.io.*;
public class HelloServlet extends HttpServlet {  1 usage
    @Override
    public void init() { System.out.println("Servlet is initialized"); }
    @Override  no usages
    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws IOException {
        response.setContentType("text/plain");
        response.getWriter().println("Welcome to my assignment servlet !");
    }
    @Override  no usages
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws IOException {
        response.setContentType("text/plain");
        response.setCharacterEncoding("UTF-8");
        PrintWriter out = response.getWriter();
        out.println("Welcome to my assignment Servlet created using xml");
        try {
            String uploadDir = "C:/Users/Axeno/IdeaProjects/myServletAssignment/src/main/resources";
            File dir = new File(uploadDir);
            if (!dir.exists()) {
                dir.mkdirs();
            }
            String fileName = "File_from_Postman.bin";
            File file = new File( pathname: uploadDir + File.separator + fileName);
            try (InputStream input = request.getInputStream();
                 FileOutputStream output = new FileOutputStream(file)) {
                input.transferTo(output);
            }
            out.println("File saved successfully!");
            out.println("Saved as: " + file.getAbsolutePath());

        } catch (Exception e) {
            out.println("File upload failed: " + e.getMessage());
        }
    }
    public void destroy() { System.out.println("Servlet Destroyed"); }
}
```

Output

```
"C:\Program Files\Java\jdk-11\bin\java.exe" ...
HTTP Status: 200
Response Body:
File uploaded successfully!
Saved as: C:\Users\Axeno\IdeaProjects\myServletAssignment\src\main\resources\uploads\20260127104621_sample (1).json
```

File saved successfuly message

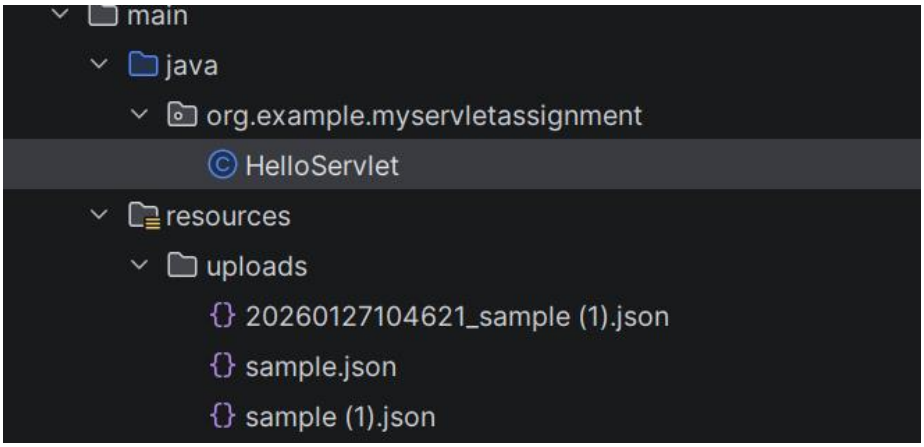
Submitted to Siddhant Rajput 27-01-2025

Body Cookies Headers (5) Test Results (1/1) 200 OK 63 ms 339 B Save Response

Raw Preview Visualize

```
Welcome to my assignment Servlet created using xml
File saved successfully!
Saved as: C:\Users\Axeno\IdeaProjects\myServletAssignment\src\main\resources\uploaded-file.bin
```

File saved in resources inside a uploads folder



Uploaded json file

pom.xml (myServletAssignment) uploaded-file.bin File_from_Postman.bin HelloServlet.java sample (1).jso


```
{
  "ordersByDate": [
    {
      "date": "2026-01-18",
      "orders": [
        {
          "orderId": "ORD3001",
          "customerName": "Amit",
          "paymentStatus": "SUCCESS",
          "products": [
            { "name": "Mobile", "quantity": 1, "price": 15000 }
          ]
        },
        {
          "orderId": "ORD3002",
          "customerName": "Neha",
          "paymentStatus": "FAILED",
          "products": [
            { "name": "Laptop", "quantity": 1, "price": 62000 }
          ]
        }
      ]
    }
```


Submitted by Rashi Sharma

Submitted to Siddhant Rajput 27-01-2025

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